

E-commerce EDA scenario challenge



Data Science Challenge

You've just joined RetailIQ as a junior data scientist.

The head of analytics has dropped a 500,000-row e-commerce dataset on your desk. Before any model gets built, she wants answers — and the data isn't clean.

Your mission

Explore the dataset, surface patterns & anomalies, then engineer features that will power a **return prediction model**. She wants your findings as insights, not just code.

Dataset in hand

500,000 orders · 30 columns · missing values · skewed prices · mixed types · ordinal loyalty tiers · high-cardinality cities

Pick a challenge to start

Phase 1

The manager says "audit the data first." Where do you even begin?

EDA

Phase 2

Someone suspects the pricing data is heavily skewed — prove it or disprove it.

Visualization

Phase 3

Which categorical columns will break your ML pipeline if left untouched?

EDA

Phase 4

Why are customers returning products? Find the patterns that matter to the business.

Scenario EDA

Phase 5

Build the full ColumnTransformer pipeline — different scalers for different columns, handle missings, encode categoricals.

Feature Engineering

Phase 6

Are any features too correlated with each other? Which ones actually predict returns?

